

HUNTER MATHIAS GILL

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LinkedIn: <https://www.linkedin.com/in/hunter-gill-4bb16532a/> | GitHub: <https://github.com/hmgill>

HuggingFace: <https://huggingface.co/hmgill> | Portfolio: <https://hmgill.github.io/>

Google Scholar: <https://scholar.google.com/citations?user=geEx8o8AAAAJ&hl=en>

Overview

AI Engineer & Applied Scientist with 6 years of experience developing multimodal AI systems, LLM agents, and computer vision models for healthcare and regulated environments. Expertise in agentic workflows, MCP integrations, RAG systems, and explainable AI. Proven record of deploying production-ready AI applications using GPT, Gemini, FastAPI, Vertex AI, Modal, and cloud-native infrastructure.

ORCID : 0000-0003-4242-7718

WORK EXPERIENCE

Contractor, Mercor

May 2026 – Present

- Roles: **Generalist, Computer Science Expert**
- Designed quality assessment rubrics for client image generation models, maintaining more than 90% billable utilization over course of projects

Research Fellow, Indiana University Indianapolis

August 2019 – October 2025

- Built REST-based vision loss simulator with FastAPI | <https://vf-simulator-22drbvsetq-uc.a.run.app/api/docs>
- Designed CAMVAS blood vessel segmentation model, pretrained on 1,000 blood vessel images. 99% accuracy, 86% IoU (intersection-over-union) score. Computed 25 vascular descriptors for proprietary chorioallantoic membrane images to perform *C. infortunatum* extract bioactivity profiling | <https://github.com/hmgill/camvas/>
- Trained RetinaRadar model on 20,000 images to assess quality (clarity, artifacts, field-of-view, usability) of standard, widefield, and ultrawidefield retina fundus photos | <https://huggingface.co/spaces/hmgill/retinaradar-demo>
- Developed BigEye by annotating and training on 1,000 retina fundus images, achieving 83% stage-specific accuracy in DR stage detection. Added SHAP analysis for extra model explainability | <https://github.com/Janga-Lab/BigEye>
- Provided technical mentoring and design documentation for a team of 5 Masters students to develop a braces detection model. Trained on multi-source dataset of 4,000 dental panoramic radiographs, with accuracy of 95% and IoU score of 83%

SELECTED PROJECTS

- Designed sandboxed GPT 5.5 agent for assessing chest x-rays (CXR). MCP connections to NVIDIA CXR-Reason and LocateAnything models for narrative writing and finding localization | <https://huggingface.co/spaces/hmgill/CXR-Agent>
- Built GPT 5.4 agent for interpreting Optical Coherence Tomography (OCT) scans, with MCP connections for specialist LO-VLM and MIRAGE foundation models. Agent uses a sandbox environment to provide custom visualizations | <https://huggingface.co/spaces/hmgill/OCT-Agent>
- Created and deployed Gemini 3 multi-agent system to segment cellular structures with SAM3, with GraphRAG-powered parameters selection. 75% faster per image compared to manual annotation for GinkGo GDPx3 dataset | <https://huggingface.co/spaces/hmgill/Cellemetry>
- Deployed and managed low-latency MCP servers with Modal serverless GPU backends for retinal fundus analysis | cup-disc: <https://fundus-cup-disc-seg.fastmcp.app/mcp>, artery/vein, fovea: <https://fundus-vascx.fastmcp.app/mcp>

Education

Ph.D. in Bioinformatics, Minor in Applied Data Science

2019 – 2025

Luddy School of Informatics, Computing and Engineering at Indiana University Indianapolis

Publications

- **H.M. Gill**, D.S. Hassan, O.B. Omoru, F.D. Bogan, J.X. Liu, A.R. Hajrasouliha, and S.C. Janga. “BigEye: a Clinically Interpretable Deep Learning Framework for Diabetic Retinopathy Detection and Stage Prediction”. *Scientific Reports*. March 2026. doi: 10.1038/s41598-026-43573-x
 - Link: <https://pubmed.ncbi.nlm.nih.gov/41796249/>
- **H.M. Gill**, Q. Mir, R. Srivastava, and S.C. Janga. “RAZOR: a Database of PCR Primers Targeting Human Respiratory Viruses”. *Nucleic Acids Research Genomics and Bioinformatics*. Jan 2026. doi: 10.1093/nargab/lqag005
 - Link: <https://pubmed.ncbi.nlm.nih.gov/41608735/>
- S. Bhatnagar, **H.M. Gill**, S.N. Arava, P.K. Singh, N. Srivastava, N. Mangir, and S.C. Janga. “Evaluating the Antiangiogenic Potential of *Clerodendrum infortunatum* Linn. through Chorioallantoic Membrane Image Analysis”. *Pharmaceutical Research – Natural Products*. Dec 2025. doi: 10.1016/j.prenap.2025.100476
 - Link: <https://www.sciencedirect.com/science/article/pii/S2950199725003362>
- D. Hassan, **H.M. Gill**, M. Happe, A. D. Bhatwadekar, A. R. Hajrasouliha, and S. C. Janga, “Combining transfer learning with retinal lesion features for accurate detection of diabetic retinopathy.” *Front Med (Lausanne)*, vol. 9, p. 1050436, Nov. 2022, doi: 10.3389/fmed.2022.1050436.
 - Link: <https://pubmed.ncbi.nlm.nih.gov/36425113/>
- N. Jha et al., “Geographical Landscape and Transmission Dynamics of SARS-CoV-2 Variants Across India: A Longitudinal Perspective.” *Front. Genet.*, vol. 12, p. 753648, Dec. 2021, doi: 10.3389/fgene.2021.753648.
 - Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8719586/>
- M. Srivastava, D. Hall, O. B. Omoru, **H.M. Gill**, S. Smith, and S. C. Janga, “Mutational Landscape and Interaction of SARS-CoV-2 with Host Cellular Components.” *Microorganisms*, vol. 9, no. 9, Aug. 2021, doi: 10.3390/microorganisms9091794.
 - Link: <https://pubmed.ncbi.nlm.nih.gov/34576690/>
- E.C. Rath, **H.M. Gill**, and Y. Bai, “Identification of potential antimicrobials against *Salmonella typhimurium* and *Listeria monocytogenes* using Quantitative Structure-Activity Relation modeling.” *PLoS ONE*, vol. 12, no. 12, p. e0189580, Dec. 2017, doi: 10.1371/journal.pone.0189580.
 - Link: <https://pubmed.ncbi.nlm.nih.gov/29236763/>

Patent Applications

S.C. Janga, H.M. Gill, Q. Mir. 2021. METHODS TO DETECT A VIRUS IN A BIOLOGICAL SAMPLE. U.S. Patent 17/997,595, filed 10/31/2022. Patent pending.

Skills

MACHINE LEARNING, LLMs, AGENTS

Frameworks (Tensorflow, Pytorch, Huggingface Transformers, sklearn), **LLMs & VLMs** (Gemma, Gemini, Claude, Llama, GLM, GPT-OSS), **Agent Frameworks** (Google ADK, smolagents, langchain, Pydantic AI, OpenAI Agents SDK), **Vector Databases** (Pinecone, chromadb), **Knowledge Graphs** (Neo4J), **RAG** (langchain), **Prompt Versioning & Observability** (Langfuse, PromptLayer), **Reproducible Workflows** (MLflow), **Data Models** (pydantic), **Traditional ML and Deep Learning** (SVM, KNN, CNN, image

classification, image segmentation, transfer learning, computer vision, ensemble learning, RF, LGBM, XGBoost, feature engineering, data cleaning, numpy, pandas)

Programming Languages, Computing Systems, Containerization

Python, R, Linux, Javascript, HTML, CSS, JSON, jQuery, AJAX, XML, PHP, SQL (stored procedures, triggers), T-SQL, MySQL Server, Siteimprove, DAX, APIs, HPC, Slurm, SFTP, Pyspark, GCP, Vertex AI, FastAPI, NextFlow, Dask, Git, GitHub CI/CD, Apptainer, Docker, Modal

Data Visualization, Processing, Annotation

Visualization (plotly, matplotlib, seaborn, Gradio, Streamlit, RShiny, Figma, Biorender), **Processing** (Databricks, ELT), **Annotation:** (Label Studio, Roboflow), Tableau, Power BI